

The Race Between Machine and Man

Tasks, displacement, reinstatement, and what Lean actually checks

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24 September 2026

Repository: `github.com/gsaco/ai-06-restrepo`

Why the unit of analysis changes

This paper asks how automation and new tasks jointly affect **aggregate** output, wages, factor shares, and employment.

- ▶ Earlier term papers center on a decision by an individual agent.
- ▶ Here the outcome comes from equilibrium across **firms, households, capital markets, and innovation**.
- ▶ A wage response cannot be read from one task: displaced workers move into the remaining labor tasks, while output and capital adjust.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Three decisions form the economic problem

Firms

Choose the lower unit cost in each feasible task; competitive prices equal costs.

Household

Chooses consumption, labor supply, and in the dynamic model saving.

Scientists

Choose research into automation or new tasks from expected innovation values.

The equilibrium allocation of tasks is the link between technology and factor prices.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

A moving unit interval of tasks

$$Y = \tilde{B} \left(\int_{N-1}^N y(i)^{(\sigma-1)/\sigma} di \right)^{\sigma/(\sigma-1)}, \quad \sigma > 0.$$



- ▶ Automation moves the feasible frontier I right; new tasks move N right and replace the lowest tasks.
- ▶ The measure of tasks stays one. The **composition** of that measure changes. At $\sigma = 1$, use the continuous CES limit.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

The threshold is economic as well as technological

$$p(i) = \begin{cases} \min\{R, W/\gamma(i)\}, & i \leq I, \\ W/\gamma(i), & i > I, \end{cases} \quad \frac{W}{R} = \gamma(\tilde{I}), \quad I^* = \min\{I, \tilde{I}\}.$$

- ▶ I says where capital *can* perform a task; $\tilde{I}$ says where it is *cost effective*.
- ▶ Strictly increasing $\gamma(i)$ gives labor comparative advantage at higher task indices.
- ▶ If $I < \tilde{I}$, technology binds. If $\tilde{I} < I$, extra automation capacity has no immediate allocation effect.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Two forces, two moving margins

Displacement: $I \uparrow$

Capital takes tasks previously done by labor. Workers crowd into fewer tasks; relative labor demand and the labor share fall when the technology constraint binds.

Reinstatement: $N \uparrow$

New, more complex tasks favor labor. The labor task range expands; relative demand, the labor share, and employment rise under the static assumptions.

Both changes can raise productivity; the distributional signs differ.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Static comparative statics need the conditions

Proposition 2: Assumptions 1–3; $\varepsilon_L > 0$; $\Lambda_I, \Lambda_N > 0$; K fixed.

Regime and change	W/R	labor share	employment
$I^* = I < \tilde{I}, dI > 0$	–	–	–
$I^* = \tilde{I} < I, dI > 0$	0	0	0
Either regime, $dN > 0$	+	+	+

$$\left. \frac{\partial \ln(W/R)}{\partial I} \right|_{I^*=I} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0, \quad \left. \frac{\partial \ln(W/R)}{\partial N} \right|_{I^*=I} = \frac{\Lambda_N}{\hat{\sigma} + \varepsilon_L} > 0.$$

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Does automation necessarily lower wages? No

Proposition 3, under Assumptions 1–3 and the **binding** threshold $I^* = I < \tilde{I}$:

$$\frac{\partial \ln W}{\partial I} = \underbrace{\left. \frac{\partial \ln Y}{\partial I} \right|_{K,L}}_{P_I \text{ productivity gain}} - \underbrace{(1 - s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L}}_{D_I \text{ displacement}}.$$

$$\boxed{\frac{\partial \ln W}{\partial I} > 0 \iff P_I > D_I}$$

- ▶ The paper states a capital threshold: for sufficiently low K (within Assumption 3), productivity dominates; at higher admissible K , displacement dominates.
- ▶ If $I^* = \tilde{I} < I$, raising feasible I has **zero** marginal effect.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

An uncalibrated numerical sign check

Hold $s_L = 0.6$, $\Lambda_I = 1$, $\hat{\sigma} = 1$, $\varepsilon_L = 1$. Then $D_I = (1 - 0.6)/(1 + 1) = 0.20$.

Illustrative P_I	$P_I - D_I$	Wage response
0.05	-0.15	falls
0.20	0	unchanged
0.35	+0.15	rises

These are **algebra checks**, not estimated model parameters. In all three binding cases, the relative wage W/R and labor share still fall under Proposition 2.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Long-run effects are different

With exogenous technology and capital accumulation (Propositions 4–5), impose $\gamma(i) = e^{Ai}$ with $A > 0$ (Assumption 1') and Assumption 2. On an interior BGP, $n = N - I \in (0, 1)$ and $R = \rho + \delta + \theta g$.

- ▶ If $n > \bar{n}(\rho)$ and automation permanently reduces n , long-run employment and the labor share fall, while the **long-run wage rises** as capital adjusts.
- ▶ If $n < \bar{n}(\rho)$, small increases in the automation frontier leave the equilibrium task allocation and effective wages unchanged.
- ▶ The short-run wage may fall even when the long-run wage rises.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Endogenous innovation: the main BGP result

Proposition 6 requires Assumptions 1' ($\gamma = e^{Ai}$, $A > 0$), 2 ($\eta \rightarrow 0$ or $\zeta = 1$), and 4 ($\hat{\sigma} > \zeta$), plus sufficiently small scientist supply $S < \bar{S}$.

Condition	Balanced growth outcome
$\rho < \bar{\rho}$	A full-automation BGP exists: $n = 0$, all tasks capital performed.
$\rho > \bar{\rho}$, $\kappa_N/\kappa_I > \bar{\kappa}$	Unique interior BGP: $n \in (\bar{n}(\rho), 1)$ and $\kappa_N v_N(n) = \kappa_I v_I(n)$.
$\rho > \bar{\rho}$, $\bar{\kappa} > \kappa_N/\kappa_I > \underline{\kappa}$	Multiple BGPs.
$\rho > \bar{\rho}$, $\kappa_N/\kappa_I < \underline{\kappa}$	Unique no-automation BGP: $n = 1$; all tasks use labor.

Here $\bar{\rho} = B - \delta - \theta g$ on a BGP; $\underline{\kappa}, \bar{\kappa}$ are the paper's innovation-ratio cutoffs. The equalities are separate boundary cases.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Why the interior path can be stable

In Proposition 6's unique interior case, $\kappa_I v_I(n) = \kappa_N v_N(n)$ with $n = N - I$. A fall in n means automation has run ahead.

- ▶ A smaller n makes producing with labor cheaper at the frontier, reducing the payoff to *further* automation and supporting new task creation.
- ▶ Stability additionally needs the relative value functions to cross with the correct slope; the paper's sufficient condition is $\kappa_N/\kappa_I > \bar{\kappa}$ and $S < \bar{S}$.
- ▶ If $\theta = 0$, equilibrium is globally saddle-path stable. If $\theta > 0$, uniqueness and asymptotic saddle-path stability are local.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Lean slide: the cost threshold

Original claim around equations (5)–(6):

$$\frac{W}{R} = \gamma(\tilde{l}), \quad l^* = \min\{l, \tilde{l}\}, \quad i < l^* \Rightarrow R < \frac{W}{\gamma(i)}.$$

```
def thresholdCostRankingSpec : Prop :=
  ∀ (γ : ℝ → ℝ) (w R I costThreshold : ℝ),
    StrictMono γ → (∀ i, 0 < γ i) → 0 < w → 0 < R →
      w / R = γ costThreshold →
        (∀ i, i < implementedThreshold I costThreshold →
          i ≤ I ∧ R < laborUnitCost w γ i) ∧
```

```
have hγlt : γ i < γ costThreshold := hmono hiCost
have hmul : γ i * R < w := by
  apply (lt_div_iff0 hR).mp
  rw [heq]
  exact hγlt
```

Lean assumes positive, strictly increasing γ on all $\mathbb{R}$, stronger than the active task interval. `hmono hiCost` transfers task order to γ ; `lt_div_iff_0 hR` uses $R > 0$. **Lean checks conditional cost ranking, not equilibrium existence.**

Build passes; no sorry.

Lean slide: the wage sign and its boundary

Proposition 3's constrained-case equation with $dN = 0$ becomes:

$$\partial_I \ln W = P_I - (1 - s_L) \Lambda_I / (\hat{\sigma} + \varepsilon_L).$$

```
noncomputable def automationWageResponse
  (productivityGain laborShare  $\Lambda_I$   $\sigma_{\text{hat}}$   $\varepsilon_L$  :  $\mathbb{R}$ ) :  $\mathbb{R}$  :=
  productivityGain - (1 - laborShare) * ( $\Lambda_I$  / ( $\sigma_{\text{hat}}$  +  $\varepsilon_L$ ))

theorem automationWageSign : automationWageSignSpec := by
  intro productivityGain laborShare  $\Lambda_I$   $\sigma_{\text{hat}}$   $\varepsilon_L$  _ _ _ _
  unfold automationWageResponse
  constructor <;> intro h <;> linarith
```

The final tactic proves the *sign equivalence* once the response equation is supplied. It does not derive the equation or P_I from production and market clearing. This is a **partial formalization** of Proposition 3. The paper-scoped fast check passed.

Where I checked the AI myself

The trap prompt was: *“Does automation necessarily reduce wages and the labour share in this model?”*

My derivation to photograph

$$\begin{aligned}dN &= 0, \quad dI > 0, \quad I^* = I < \tilde{I}, \\ \partial_I \ln W &= P_I - D_I, \\ D_I &= (1 - s_L)\Lambda_I/(\hat{\sigma} + \varepsilon_L), \\ \partial_I \ln W > 0 &\iff P_I > D_I.\end{aligned}$$

Hand derivation photo pending
Place at hand/derivation.jpg

Verdict: the flat claim “automation lowers both” is false for wages; and even the labor-share decrease requires an effective, binding expansion of automated tasks.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

What I did and what remains open

Analytical and computational work

- ▶ Reconstructed the technology and cost thresholds.
- ▶ Derived the wage sign condition.
- ▶ Tested both signs with illustrative numbers.
- ▶ Ran two Lean proofs and the paper-scoped fast check.

Translation boundary

- ▶ No Lean proof of equilibrium existence or uniqueness.
- ▶ No derivation of P_I , Λ_I , or the balanced growth conditions from primitives.
- ▶ Full source-to-Lean semantic audit remains open; the generated folder records that status.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.

Takeaway

Automation reallocates tasks to capital. New tasks reinstate labor at the frontier.

- ▶ Under the static model's assumptions, binding automation lowers relative wages, the labor share, and employment.
- ▶ The **absolute wage** rises only when the productivity gain exceeds displacement; in the long run, capital adjustment can reverse a short-run wage decline.
- ▶ Lean made the threshold, positive denominators, and proof boundary explicit. It verified a useful slice of the argument, not the whole equilibrium model.

Source: Acemoglu and Restrepo, NBER WP 22252, revised June 2017.